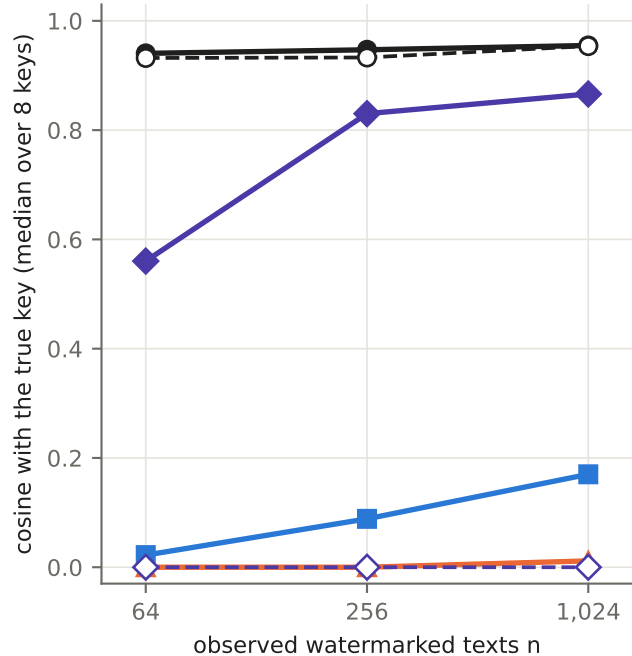
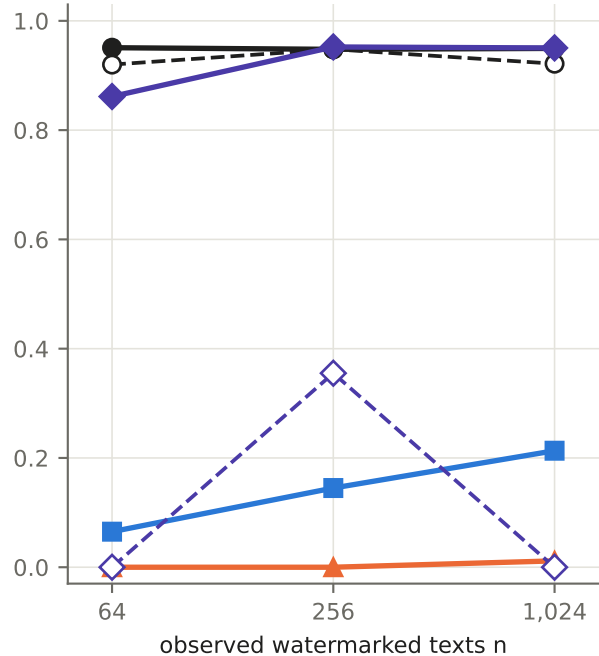


$\rho = 0.35$: key recovery (study keys)

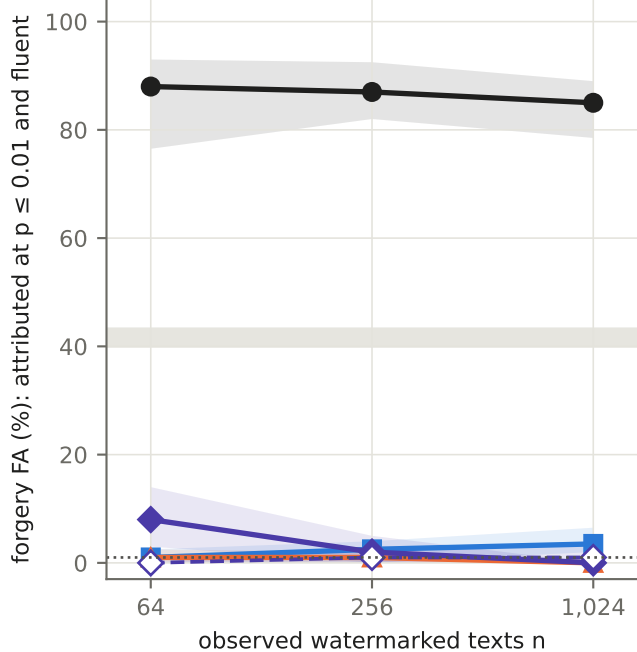


$\rho = 0.50$: key recovery (study keys)

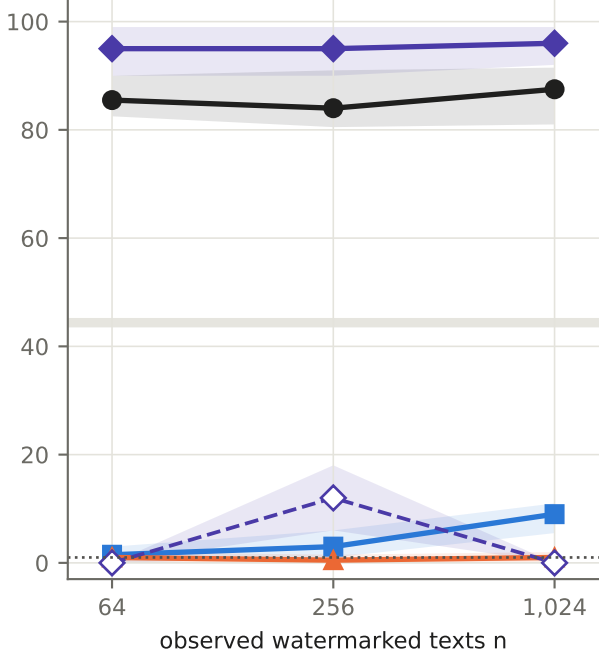


- fixed key, Route A', known layer
- fixed key, Route A', layer search
- context-hashed, h = 1: count-weighted cos
- context-hashed, h = 4: count-weighted cos
- rotation: clustering attacker (median cluster)
- rotation: naive attacker (best key)

$\rho = 0.35$: forgery acceptance (study keys)



$\rho = 0.50$: forgery acceptance (study keys)



- fixed key
- context-hashed, h = 1
- context-hashed, h = 4
- rotation, clustering attacker
- rotation, naive attacker
- bar: half of each arm's genuine FA (40-43%)
- random-key / random-secret controls (max 1%)